

Approach A+

The Instrumented Studio inside ChatGPT Edu

Everything from Approach A, plus a crew of sub-agents inside every advisor and a live insights spreadsheet: auto-grading, session summaries, and per-student usage analytics still nothing you host, still ~\$0 per student.

Teal box = lives inside ChatGPT Edu. Green box = lives in IU Google Workspace (serverless). The small pills inside each advisor are its sub-agents; orange pills are the common crew embedded in all four advisors.

Flow Notes	How information moves through the architecture
1	student checks in (name / roster ID) every session starts identified
2	the sub-agents use what's built into ChatGPT plus one new bridge
3	structured JSON events flow out on every milestone
4	at the end of each stage the studio produces...
5	submitted to the gradebook

1. The people who use it

Component	Description	Technical Context
Student	Opens the shared workspace, checks in with their roster ID, and runs a real project through four stages. Drafts first; the AI challenges; every milestone is logged.	
Instructor (you)	Builds the workspace once then watches a live dashboard instead of grading cold. Moderates AI-suggested grades; drills into any student's tab.	

2. The studio four advisors, each with an embedded sub-agent crew

PM Studio = one ChatGPT Project + one orchestrating GPT + sub-agent crews

Component	Description	Technical Context
Scope Advisor	Charter, work breakdown, scope in / out. Embedded roles: Insights Agent, Auto-Grader, Summarizer, WBS Decomposer, Assumption Auditor	
Resource & Cost Advisor	Labor hours, costs, resource conflicts. Embedded roles: Insights Agent, Auto-Grader, Summarizer, Cost Engine (deterministic), Wage-Data Agent, Conflict Checker	
Risk Advisor	Finds risks; quantifies them with simulation. Embedded roles: Insights Agent, Auto-Grader, Summarizer, Monte Carlo Sim Agent, Risk-Register Keeper	
Stakeholder Advisor	Role-plays officials, community, funders. Embedded roles: Insights Agent, Auto-Grader, Summarizer, Persona Player, Negotiation Debrief Coach	
Review Board	Stage-gate before each advance. Pulls the Auto-Grader's rubric scores, pushes back Socratically, and won't pass thin work.	stage-gate. grades logged to the spreadsheet at every gate
Living Project File ChatGPT Canvas ("Studio Doc")	One side-by-side document every advisor reads and updates, so each mode knows what the last one decided. Continuity without copy-paste.	ChatGPT Canvas document. NOT the Canvas LMS

3. Tools the crews run on

Component	Description	Technical Context
Simulation & math runtime	Runs the versioned montecarlo.py and cost roll-ups as real code the LLM never does the math.	Code Interpreter. script shipped as a knowledge file
Reference files	Rubric, templates, persona briefs, cost snapshot, and the Monte Carlo script itself.	knowledge files (versioned in the build repo)
Logging Action (NEW)	After every milestone (draft, critique, sim run, gate, grade) the GPT posts a structured event to the spreadsheet.	GPT Action. OpenAPI schema. shared-token auth

4. The insights spreadsheet (one workbook for the whole class)

Component	Description	Technical Context
Event receiver	A tiny script that catches each event, stamps it, and files it creating a new student tab automatically on first contact.	Google Apps Script web app (doPost). no server
Master Dashboard tab (the standard sheet)	One row per student: current stage, per-advisor grades, sessions, critique-depth, AI-reliance index, latest summary, flags.	rollup formulas over the event log
One tab per student	Named with the student's name. Full chronology: what they drafted, how they pushed back, sim results, gate outcomes, grades, insight notes.	auto-created. timestamp advisor sub-agent event score note

5. What the student turns in

Component	Description	Technical Context
Competency report (AI-generated, session-stamped)	Draft + justifications + AI critique + rubric score, now footer-stamped with a session ID so you can cross-check it against the student's tab. Tamper-evident, not tamper-proof.	session ID ties the PDF to the spreadsheet audit trail

6. Submission & grading (your existing tools)

Component	Description	Technical Context
Canvas LMS	Students submit the report as a normal assignment. Canvas stays the official gradebook.	Instructure GPT auto-grades are advisory, We decide the grade
Dashboard-assisted review	Open the Master Dashboard, sort by flags, moderate the AI-suggested grades, and spot-check reports against student tabs.	minutes per student instead of a cold read

Glossary

Term	Definition
Sub-agent	A specialist role the orchestrating GPT switches into for one job (grading, summarizing, simulating). One model, many hats orchestrated by protocol, not parallel processes.
Insights Agent	The interactive data agent embedded in all four advisors. Silently logs usage patterns; students and instructors can also ask it directly ("How am I using AI? Where am I over-relying?").
Auto-Grader	Sub-agent that scores work against your rubric at each stage gate and logs score + rationale. Advisory the instructor moderates.
Summarizer	Sub-agent that writes a 3-line digest per session: what the student is working on, decided, and stuck on. Feeds the dashboard.
AI-reliance index	Ratio of AI content accepted verbatim vs. challenged / modified by the student. High reliance + low critique depth = a teaching moment.
Critique depth	0-3 rating of how substantively the student pushed back on AI drafts in a session.
GPT Action	A Custom GPT's ability to call an external API, defined by an OpenAPI schema. Here: the only bridge out of ChatGPT, used for logging.
Apps Script	Google's built-in scripting for Sheets. Deployed as a web app it becomes a free, serverless webhook no hosting.
Session ID	A unique stamp on each report footer and each logged event, tying submissions to the audit trail.
Code Interpreter	ChatGPT's built-in Python runtime runs the versioned Monte Carlo script and cost math deterministically.
Codex	OpenAI's coding agent (CLI / IDE / cloud), included with ChatGPT Edu sign-in. Builds and maintains everything in the repo: prompts, Apps Script, schemas, tests.
Codex Skill	A reusable SKILL.md instruction pack Codex loads on demand. Each build task below is captured as a skill so next term's rebuild is a re-run, not a rewrite.
ChatGPT Canvas	The AI-edited side-by-side document holding the Living Project File. (Not the Canvas LMS.)